BO CDA 203	Design of Algorithms

Course Description:

Model of Computations: RAM Model of computation, uniform cost model, logarithmic cost model. Complexity Analysis: Big O, omega, theta notations, solving recurrence relation. Data Structure: binary search trees, AVL trees and red-black trees, B-trees, hashing, Priority queues, Heaps. Sorting algorithms: Merge sort, Quick sort, Heap sort, Randomized quick sort, Lower bound of comparison based sorting, Counting sort, Radix sort, Bucket sort. Algorithm design techniques: Greedy, Divide and Conquer, Dynamic Programming. Graph Algorithms: BFS and DFS, Minimum spanning trees- Kruskal and Prim algorithm, Shortest Path- Dijkstra, Bellman-Ford, Johnson algorithm, Network flow- Ford-Fulkerson algorithm. NP-Completeness: Class P, NP, NP-hard and NP-complete, Examples of NP-complete problems.

Practical component: Lab to be conducted on a 2-hour slot weekly. It will be conducted with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should be synchronized with the laboratory classes by using C/Python. The problems to be solved should involve all the algorithm designing techniques that are covered in the theory class.

Learning Resources:

1. M. A. Weiss, Data Structures and Algorithms in C++, 4th Edition, Pearson, 2014
2. J. Kleinberg and Eva Tardos, Algorithm Design, Pearson Education, 2013
3. T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, Introduction to Algorithms, MIT Press, 2009
4. A. Aho, J. E. Hopcroft and J. D. Ullman, The Design and Analysis of Computer Algorithms, Pearson, 1974